

Researchers develop ultra-strong metal 3D printing alloy

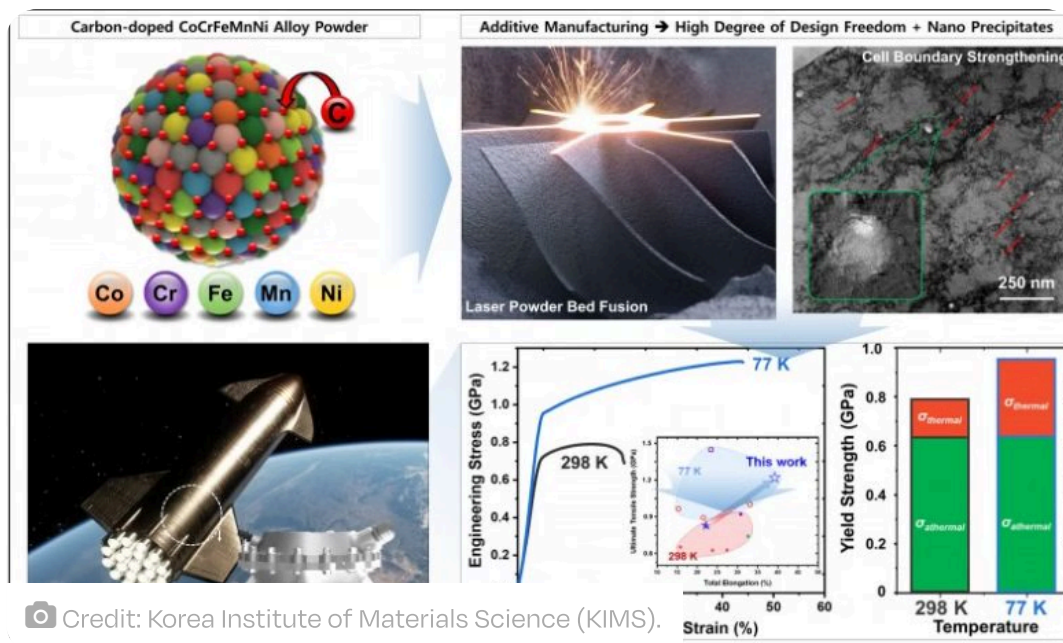
KIMS developed the material for space environments in collaboration with Gyeongsang National University and POSTECH



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A research team led by Dr. Jeong Min Park from the Nano Materials Research Division at the Korea Institute of Materials Science (KIMS), in collaboration with Professor Jung Gi Kim of Gyeongsang National University and Professor Hyoung Seop Kim of [Pohang University of Science and Technology \(POSTECH\)](#), has successfully developed a new high-performance metal 3D printed alloy tailored for space environments. The newly developed alloy demonstrated exceptional mechanical performance at extreme cryogenic temperatures as low as -196°C – proving its [potential for space exploration](#) and extreme environment applications.

The research team added a small amount of carbon to CoCrFeMnNi alloy, which exhibits excellent properties in cryogenic conditions. This alloy powder was then processed using Laser Powder Bed Fusion

(LPBF). The technology allows for the maximization of the strengthening effect of carbon addition to the alloy via finely distributed nano-carbides at the boundaries of nano-sized cell structures. As a result, the team achieved a combination of tensile strength (the ability to resist forces) and ductility (the ability to endure deformation before failure) that was over 140% better than carbon-free alloys in cryogenic environments. In particular, the elongation of the alloy is twice as high at 77K compared to 298K. This technology also offers a potential guideline for alloying design in AM to produce high-performance products with excellent load-bearing capacity for use in cryogenic applications. Another key distinction of this technology is its ability to fine-controlling microstructure through AM.

This metal 3D printing alloy can be applied to complex components such as injectors that spray fuel in [space exploration rockets](#), and turbine nozzles that extract energy. It enhances the performance and extends the lifespan of parts used in space and other extreme environments. Moreover, it overcomes the limitations of low-temperature toughness in existing 3D printed alloys.

"This research presents a significant breakthrough in developing new alloys for extreme environments, offering new possibilities. Through 3D printing technology that surpasses the manufacturing limits of [conventional space exploration components](#), we can significantly improve the performance of parts used in space launch vehicles," said Dr. Jeong Min Park, the senior researcher and project leader.

This research was funded by the fundamental project of KIMS ('Development of design for additive manufacturing to develop superhard heterogeneous materials with complex design' and 'Development of High Performance Materials and Processes for Metal 3D Printing'). [The research findings were published](#) in the journal 'Additive Manufacturing (IF: 11)'. The research team plans to continue further studies to enhance the commercialization potential of this technology and conduct additional research to verify its performance in extreme environments.